

Compositional, tree-aware VAEs in biomevae

Model diagrams and mathematics: PhILR-VAE, TreeDTM-VAE, CAPDA-VAE

biomevae

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Outline

1. Setup: compositional microbiome data
2. PhILR-VAE
3. TreeDTM-VAE
4. CAPDA-VAE

The data lives on the simplex

MetaPhlAn / SGB tables give, per sample, non-negative counts $\mathbf{x} \in \mathbb{N}_0^p$ or relative abundances $\mathbf{c} \in \Delta^{p-1}$ with

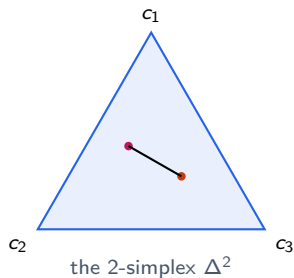
$$\Delta^{p-1} = \left\{ \mathbf{c} \in \mathbb{R}_{>0}^p : \sum_i c_i = 1 \right\}.$$

A pseudocount $\alpha > 0$ closes counts to a composition:

$$c_i = \frac{x_i + \alpha}{\sum_j (x_j + \alpha)}.$$

Only ratios carry information. Euclidean distance / MSE on $\mathbf{c}$ is geometrically wrong; the correct metric is the **Aitchison distance**

$$d_A(\mathbf{c}, \mathbf{c}') = \left\| \text{clr}(\mathbf{c}) - \text{clr}(\mathbf{c}') \right\|_2, \quad \text{clr}(\mathbf{c})_i = \log \frac{c_i}{(\prod_j c_j)^{1/p}}.$$



Three model families, one taxonomy backbone:
PhILR-VAE (isometry), **TreeDTM-VAE** (tree likelihood), **CAPDA-VAE** (domain alignment).

PhILR-VAE — a VAE in isometric log-ratio space

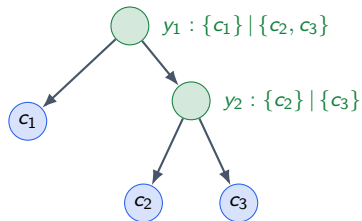
Idea. Map the simplex to flat space with a *tree-defined isometry*, run a vanilla Gaussian VAE there, and map back — so Euclidean operations on the latent respect Aitchison geometry exactly.

Sequential Binary Partition (SBP).

Each internal node of the taxonomy splits its descendant leaves into R ($|R| = r$) vs S ($|S| = s$), giving one orthonormal *balance*

$$\psi_i = \begin{cases} +\sqrt{\frac{s}{r(r+s)}} & i \in R, \\ -\sqrt{\frac{r}{s(r+s)}} & i \in S, \\ 0 & \text{otherwise.} \end{cases}$$

Stacking $p - 1$ balances gives $\Psi \in \mathbb{R}^{p \times (p-1)}$ with

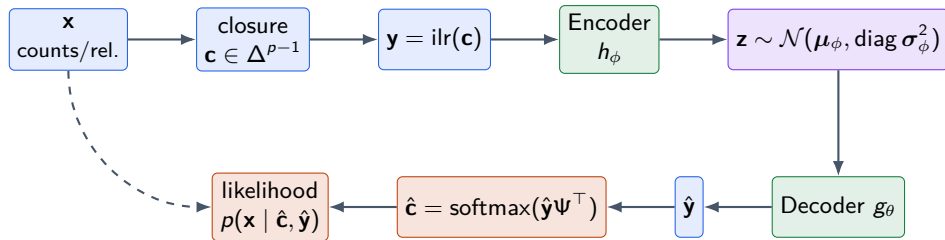


Forward / inverse transform

$$\mathbf{y} = \text{ilr}(\mathbf{c}) = \log(\mathbf{c}) \Psi \in \mathbb{R}^{p-1}, \quad \hat{\mathbf{c}} = \text{softmax}(\hat{\mathbf{y}} \Psi^\top).$$

Isometry (Aitchison $\equiv$ Euclidean)

PhILR-VAE — computation graph



Encoder $\mathbf{x} \rightarrow \mathbf{c} \rightarrow \mathbf{y} \rightarrow h_\phi(\mathbf{y}) \rightarrow (\mu_\phi, \log \sigma_\phi^2)$, with $\mathbf{z} = \mu_\phi + \sigma_\phi \odot \epsilon$.

Decoder $\mathbf{z} \rightarrow \hat{\mathbf{y}} \rightarrow \hat{\mathbf{c}} = \text{softmax}(\hat{\mathbf{y}}\Psi^\top) \in \Delta^{p-1}$ (closure restored).

Why principled

Outputs always lie in Δ^{p-1} ; Euclidean decoding in ILR space is exact on the simplex, so *no* extra geometric/consistency penalty is needed.

PhILR-VAE — likelihoods and objective

Five likelihoods share the encoder/decoder; only $\log p(\mathbf{x})$ changes:

- `philir_gaussian` (default, relative input) — logistic-normal on Δ^{p-1} : $\mathbf{y} \sim \mathcal{N}(\hat{\mathbf{y}}, \text{diag } \sigma_{\text{obs}}^2)$.
- `multinomial`: $\mathbf{x} \mid \hat{\mathbf{c}}, N \sim \text{Mult}(N, \hat{\mathbf{c}})$.
- `dirichlet_multinomial`: $\alpha = \alpha_0 \hat{\mathbf{c}}$, global overdispersion.
- `dirichlet_tree_multinomial`: per-clade DTM (shares the TreeDTM-VAE likelihood, next section).
- `dirichlet_tree`: continuous Dirichlet-tree for closed data.

ELBO objective

$$\mathcal{L} = -\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})] + \beta_t \text{KL}(q_\phi \parallel \mathcal{N})$$

with closed-form Gaussian KL

$$\text{KL} = \frac{1}{2} \sum_{j=1}^d (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2),$$

β_t a linear warm-up, $\mathcal{R}$ a small L_2 on the likelihood parameters.

TreeDTM-VAE — a likelihood at every internal node

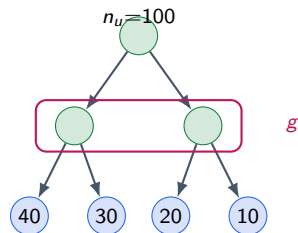
Idea. Instead of a leaf-wise likelihood, factor the count distribution over the taxonomy: an independent *local split* at every internal node, with overdispersion learned *per clade*.

Rooted tree $\mathcal{T}$ with internal nodes $\mathcal{I}$, leaves $\mathcal{L}$.
Observed vector $\mathbf{y}^{(i)} \in \mathbb{R}_{\geq 0}^N$ is aligned to *every* node: leaves are raw counts, each internal entry is the sum of its children.

Decoder produces split logits, then a masked softmax per sibling group:

$$\mathbf{a}^{(i)} = h_{\theta}(\mathbf{z}^{(i)}) \in \mathbb{R}^E, \quad E = \sum_{u \in \mathcal{I}} K_u,$$

$$\pi_{u \rightarrow v}^{(i)} = \frac{\exp(a_{u \rightarrow v}^{(i)})}{\sum_{v' \in \text{Ch}(u)} \exp(a_{u \rightarrow v'}^{(i)})}, \quad v \in \text{Ch}(u).$$



Sibling-centred log-ratio encoder

$$\ell_{u \rightarrow v}^{(i)} = \log(y_v^{(i)} + \tau) - \frac{1}{K_u} \sum_{v' \in \text{Ch}(u)} \log(y_{v'}^{(i)} + \tau)$$

gauge-fixes the within-group d.o.f.; MLP
 $g_{\phi}(\ell) \rightarrow (\mu_z, \log \sigma_z^2)$.

TreeDTM-VAE — three node likelihoods

Joint log-likelihood is always a sum over internal nodes, $\log p(\mathbf{y}^{(i)} | \mathbf{z}^{(i)}) = \sum_{u \in \mathcal{I}} \log p(\mathbf{x}_u^{(i)} | \cdot)$, with child counts $\mathbf{x}_u^{(i)} = (x_{u \rightarrow v}^{(i)})_v$ and $n_u^{(i)} = \sum_v x_{u \rightarrow v}^{(i)}$.

Tree-Multinomial (no overdispersion)

$$p(\mathbf{x}_u | \boldsymbol{\pi}_u) = \binom{n_u}{\mathbf{x}_u} \prod_{v \in \text{Ch}(u)} (\pi_{u \rightarrow v})^{x_{u \rightarrow v}}$$

Dirichlet-Tree-Multinomial (default)

Per-group concentration $\alpha_g^0 > 0$ from the decoder, $\boldsymbol{\alpha}_g^{(i)} = \alpha_g^0 \boldsymbol{\pi}_g^{(i)}$:

$$p(\mathbf{x}_u | \boldsymbol{\alpha}_g) = \binom{n_u}{\mathbf{x}_u} \frac{\Gamma(\alpha_{g,0})}{\Gamma(n_u + \alpha_{g,0})} \prod_v \frac{\Gamma(x_{u \rightarrow v} + \alpha_{g,v})}{\Gamma(\alpha_{g,v})}$$

with $\alpha_{g,0} = \sum_v \alpha_{g,v} = \alpha_g^0$. Dispersion is *per clade*.

Continuous Dirichlet-Tree (closed data)

For $\mathbf{c} \in \Delta^{L-1}$, with child fractions $\mathbf{q}_u = \text{normalise}(\mathbf{x}_u + \boldsymbol{\eta})$:

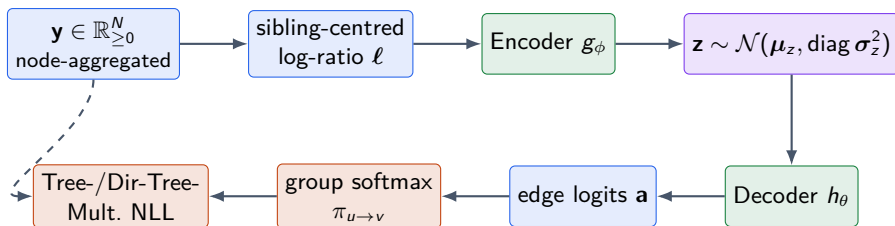
$$\begin{aligned} \log p(\mathbf{q}_u | \boldsymbol{\alpha}_g) &= \log \Gamma(\alpha_{g,0}) - \sum_v \log \Gamma(\alpha_{g,v}) \\ &\quad + \sum_v (\alpha_{g,v} - 1) \log q_{u \rightarrow v}. \end{aligned}$$

Objective

$$\mathcal{L}^{(i)} = -\log p(\mathbf{y}^{(i)} | \mathbf{z}^{(i)}) + \beta_t \text{KL}(q_\phi \| \mathcal{N}(\mathbf{0}, I)) + \gamma \|\boldsymbol{\alpha}^0\|^2,$$

$$\text{KL} = \frac{1}{2} \sum_i (\mu_{z,j}^2 + \sigma_{z,j}^2 - 1 - \log \sigma_{z,j}^2), \text{ optional free-bits}$$

TreeDTM-VAE — computation graph



Closure is enforced *by the likelihood* (each sibling group is a proper multinomial / Dirichlet-multinomial), so proportions live exactly on the simplex — statistically valid for both fixed-depth counts and pre-closed relative abundances.

CAPDA-VAE — class-conditional, adversary-free domain alignment

Idea. A supervised, taxonomy-biased VAE that removes *study* nuisance from $P(\mathbf{z} \mid y)$ *without* an adversary, by pulling together same-class latent statistics across studies — preserving the between-class signal that marginal DA-VAEs destroy under label shift.

Multi-resolution taxonomy input

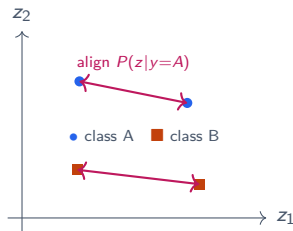
Per-species CLR *plus* abundances aggregated up the taxonomy (genus/family/order/phylum):

$$\mathbf{x} = \left[\underbrace{\text{clr}(\mathbf{X})}_{\text{species}} \mid \underbrace{\log(1 + \mathbf{A}_{g,f,o,p})}_{\text{aggregates}} \right],$$

$$\text{clr}(\mathbf{X})_i = \log(X_i + \epsilon) - \frac{1}{p} \sum_j \log(X_j + \epsilon).$$

Network

MLP encoder $\rightarrow (\boldsymbol{\mu}, \log \sigma^2)$, $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$; MLP decoder $\rightarrow \hat{\mathbf{x}}$; **class head** on the latent $\ell_{\text{cls}} = \text{cls}(\mathbf{z})$.



Same-class study means pulled to a shared per-class centroid; different classes stay apart.

CAPDA-VAE — conditional alignment (adversary-free)

For class c and study s , let the per-(study,class) latent mean be

$\mathbf{m}_{c,s} = \text{mean}\{\boldsymbol{\mu}_i : y_i = c, d_i = s\}$ and the class centroid $\bar{\mathbf{m}}_c = \text{mean}_s \mathbf{m}_{c,s}$ (over studies present).

First moment (conditional mean)

$$\mathcal{A}_1 = \frac{1}{|\mathcal{C}'|} \sum_{c \in \mathcal{C}'} \frac{1}{|\mathcal{S}_c|} \sum_{s \in \mathcal{S}_c} \|\mathbf{m}_{c,s} - \bar{\mathbf{m}}_c\|_2^2,$$

over classes $\mathcal{C}'$ seen in ≥ 2 studies.

Second moment (conditional CORAL)

$$\mathcal{A}_{\text{cov}} = \frac{1}{|\mathcal{C}''|} \sum_{c \in \mathcal{C}''} \sum_s \|\text{Cov}_{c,s} - \bar{\Sigma}_c\|_F^2,$$

with $\text{Cov}_{c,s} = \text{Cov}(\boldsymbol{\mu} : y=c, d=s)$,

Total objective

$$\begin{aligned} \mathcal{L} = & \underbrace{\|\hat{\mathbf{x}} - \mathbf{x}\|_2^2}_{\text{recon (MSE)}} + \beta_t \text{KL} + \alpha \text{CE}(\ell_{\text{cls}}, y; \mathbf{w}) \\ & + \lambda_t \mathcal{A}_1 + \frac{\lambda_t}{\gamma} \gamma_{\text{cov}} \mathcal{A}_{\text{cov}}, \end{aligned}$$

CE class-weighted; β_t, λ_t linear warm-ups
($\lambda_t = \gamma \min(1, t/T_{\text{align}})$).

CAPDA-VAE — why *conditional*? the taxonomy/invariance trade-off

A formal trade-off governs taxonomy bias vs. domain invariance on the balanced two-study risk $\mathcal{E}$:

Taxonomy upper bound (T)

A taxonomy-aware backbone (PhILR / TreeDTM) with no invariance constraint attains $\mathcal{E}_{(T)} \leq \hat{\mathcal{E}} + \mathcal{O}(\sqrt{\mathfrak{C}(\mathcal{F}_T)/n})$, the variance-reduction half.

Invariance lower bound (I)

Enforcing *marginal* domain invariance (DIVA / PhyloDIVA style) costs $\mathcal{E}_{(I)} \geq \frac{\Delta_Y}{2} - \delta$, where Δ_Y is the between-study label shift. Invariance *hurts* precisely when composition shifts with the label.

CAPDA's resolution

Align *within each class* ($P(\mathbf{z} \mid y)$), not the marginal $P(\mathbf{z})$. This removes the study nuisance term without paying the $\Delta_Y/2$ penalty of (I) — and keeps the taxonomy bias (T) via the multi-resolution CLR input. Adversary-free: alignment is a closed-form moment-matching penalty, not a min-max game.

Single-study limit

With one domain, $\mathcal{A}_1 = \mathcal{A}_{\text{cov}} \equiv 0$ (any class needs ≥ 2 studies): CAPDA degrades gracefully to a supervised taxonomy-biased VAE with leak-free stratified K -fold OOF stacking.

Summary — three compositional VAEs

	PhILR-VAE	TreeDTM-VAE	CAPDA-VAE
Encoder input	ILR coords $\mathbf{y} = \log(\mathbf{c})\Psi$	sibling-centred log-ratios per node	[clr(species) log-aggr]
Tree enters via	SBP contrast basis Ψ	per-node likelihood factorisation	multi-resolution aggregation
Likelihood	logistic-normal / multinomial / DTM	Tree-/Dir-Tree-Multinomial	Gaussian recon + class head
Extra term	— (isometry is exact)	per-clade $\ \alpha^0\ ^2$	conditional align $\mathcal{A}_1, \mathcal{A}_{\text{cov}}$
Best for	metric / embedding quality	clade-specific overdispersion	cross-cohort label prediction

All three: standard Gaussian latent $\mathcal{N}(\mathbf{0}, I)$, reparameterised sampling, ELBO with linear β_t warm-up. Theory: [docs/philrvae_theory.md](#), [docs/tree_dtm_vae_theory.md](#), [docs/capda_vae_theory.md](#).